Kenny Valent Winalda Sembiring

Informatics Undergraduate · Software Engineering · AI / ML · Backend & Infrastructure

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EDUCATION

Universitas Multimedia Nusantara (UMN) 2024 – Est. Aug 2028

S1 Informatika GPA 3.53 / 4.00

- Relevant coursework: Algorithms & Data Structures, Object-Oriented Programming (Kotlin), Advanced Web Programming, Machine Learning, Computer Networking, Computer & Communications Security

EXPERIENCE

Asset Management Division — Kelompok Studi Pasar Modal 2025 – Present

UMN Student Organization · Tangerang

- Analyzed Indonesian-listed equities (TUGU) for Stock Quest 2026 competition; built full three-statement financial model for TUGU with 2026–2028 projections
- Developed valuation frameworks incorporating PSAK 117 / IFRS 17 adoption impacts on insurance sector reporting
- Produced investment memos and pitch materials for peer review and competition submission

PROJECTS

Forge — Open-Source Agentic Coding CLI 2026

github.com/KennyUMN/forge · TypeScript / Node.js · Open Source

- Co-building an open-source agentic coding CLI kernel in TypeScript (Node 20+); derived the architecture from a comparative study of 10 existing agentic coding tools (opencode, Cline, Aider, OpenHands, SWE-agent, Goose) before writing an original codebase
- Shipped the kernel phase end to end: session manager, model-provider abstraction, agent loop with permission gate, plugin/tool registry with MCP client, and a CLI with session resume — 154/154 tests passing
- Added multi-provider model routing through OpenRouter (300+ models) behind a single provider interface
- Roadmap targets native multi-agent orchestration, a plugin ecosystem, and deep local dev-tool integration (git, LSP, sandboxing)

memory-core — Shared Long-Term Memory for AI Agents 2026

github.com/KennyUMN/memory-core · Python / FastAPI / MCP · Open Source

- Built a shared memory server that Claude Code, Codex, and Antigravity read and write over MCP (six tools: remember, recall, forget, profile, timeline, search_docs) and a REST/JSON API
- Implemented hybrid recall on SQLite: FTS5 BM25 plus Qwen3 embeddings fused with Reciprocal Rank Fusion, falling back to BM25-only when the embedding service is down
- Added write hygiene: near-duplicate detection (cosine ≥ 0.86), supersede and soft-forget with reasons, and automatic redaction of API keys and tokens before storage
- Deployed on a self-hosted Ubuntu server with per-agent bearer tokens, per-agent usage logging, a web dashboard, and rotating SQLite backups

FinRAG-ID — Indonesian Financial Document Q&A 2025

github.com/KennyUMN/finrag-id

- Built a production-style local-first RAG pipeline on LlamaIndex + Qdrant + Ollama; targets Indonesian financial documents (laporan tahunan, prospektus)
- Implemented hybrid retrieval (BM25 with Sastrawi Indonesian stemmer + BGE-M3 dense embeddings) fused via Reciprocal Rank Fusion for superior recall on numeric financial data
- Runs fully offline via Ollama (Llama 3.1 8B Q5_K_M GGUF quantized), with OpenRouter as an automatic cloud fallback for higher-capability models
- Evaluated with RAGAS metrics (faithfulness, answer relevancy, context precision) + per-query latency tracking; containerized via Docker Compose for one-command deployment
- FastAPI backend with /query, /upload, /health endpoints; Streamlit UI with source citation display

Lembar Janji — Overseas Job Offer Compliance Engine Sep 2026

IFEST 2026 Hackathon (4th Place) · Next.js (App Router) + TypeScript · Fullstack

- Built a 3-tier deterministic legal compliance validation engine evaluating overseas job offer flyers against 10 statutory requirements under Law No. 18/2017 (UU PPMI)
- Engineered modular architecture decoupling vision-model extraction from core validation rules: offline P3MI agency matching, cost compliance rules, and zero-PII card rendering (399 Vitest tests passing)

Paspor Petani — Offline-First Field App for Coffee Farmers Jul 2026

GarudaHacks 7.0 · Agriculture & Food Track · React Native (Expo) + TypeScript · Android

- Ported an offline-first PWA blueprint into a native Android app in under 12 hours: plot tagging, deforestation checks, tiered verification cards, and offline-capable maps
- Implemented hash-chain record integrity with expo-crypto so field data captured without connectivity stays tamper-evident

PPE Detection System — YOLOv9c Semi-Supervised Learning 2025 – 2026

IF441 Group Project · Kaggle: kennyvws

- Trained YOLOv9c on a merged Kaggle PPE dataset; implemented a two-phase SSL pipeline: supervised baseline → pseudo-label generation → self-training
- Designed a Compliance State Machine using IoU-based Person-to-PPE mapping for structured safety event detection
- Debugged training pipeline issues (CLI vs Python API), managed multi-phase model checkpointing on Kaggle

Self-Hosted AI Homeserver 2026 – Present

Personal Infrastructure Project · Ubuntu 24.04 / CasaOS

- Running Ubuntu 24.04 + CasaOS with Dockerized services (Ollama, OpenClaw AI gateway, Tailscale VPN overlay)
- Configured OpenClaw to route Telegram/WhatsApp requests through OpenRouter (Llama, Gemini, Claude) with JSON-level config management via SSH
- Practical exposure to local LLM serving, container networking, bind-mount management, and API key rotation

Health Insurance Claims Forecasting — DSC MCF ITB Feb 2025

- Reduced MAPE from 14.13% to 6.46% using multiplicative seasonal decomposition to preserve monthly seasonality patterns
- Outperformed flat ensemble baseline by explicitly modeling claim volume cycles from 2024 historical data

TECHNICAL SKILLS

AI / LLM	Ollama, vLLM (familiar), LlamaIndex, LangChain (familiar), RAGAS, GGUF quantization, OpenRouter API, HuggingFace, Model Context Protocol (MCP)
ML / Data	PyTorch, YOLOv9c, Scikit-learn, Pandas, NumPy, Tableau, semi-supervised learning, seasonal decomposition
Infrastructure	Docker, Docker Compose, Ubuntu Server, CasaOS, Tailscale, SSH, Qdrant, ChromaDB
Languages	Python, TypeScript / JavaScript, Kotlin, SQL, Bash
Web / Mobile	Next.js, React, Tailwind CSS, React Native (Expo), FastAPI, Streamlit
Other	Node.js, Git / GitHub, REST API integration, Model Context Protocol (MCP), unit & integration testing, technical documentation

COMPETITIONS & ACTIVITIES

- IFEST 2026 — 4th Place (Juara Harapan 1), National Hackathon by Universitas Padjadjaran (Lembar Janji)
- GarudaHacks 7.0 2026 — Agriculture & Food Systems Track, Southeast Asia's largest hackathon (Paspor Petani Mobile)
- Stock Quest 2026 — Equity Research Competition (Indonesian capital markets, multi-sector analysis)
- DSC MCF ITB 2025 — Data Science Competition (health insurance claims forecasting, top MAPE improvement)
- Dinamic 2025 — Equity Research Competition (Indonesian capital markets, multi-sector analysis)